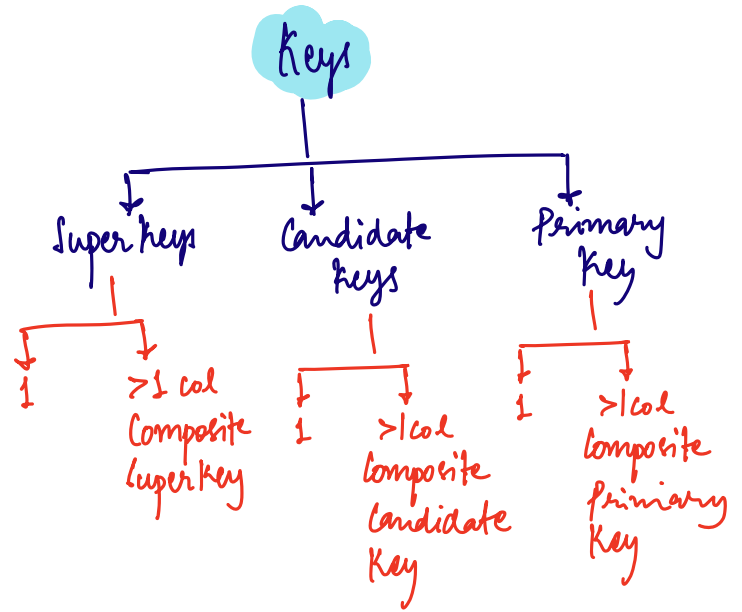


Agenda

- Foreign Keys
- What is CRUD
- Sakila Database



Foreign Keys

FK has nothing to do with your CK, PK or SK.

Students

id	name	email	phone	batch-id
	Alok	alok@	123	1
	Ojasvi	aj..	456	2
	Bhakti	bhakt@	789	2
	Sumit	sumit@	10111	1

Primary Key

Foreign Key of students table

Batches

id	name	st-date	end-date	student-count
1	Apr 22	19/8/2023	19/9/2024	100
2	March 23	19/8/2024	20/9/2025	250

Primary Key of batches table

Foreign Key:-
constraints.

A coln. or a set of columns that will help to uniquely identify exactly 1 row from another (foreign) table.

Note:- Specifying FK in your database helps ensure data integrity.

Example:-

id	name	batch-id
1	Varun	1
2	Bhakti	2
3	Sumit	2
4	Ojasvi	3
5	Mukhul	2

id	name
1	Apr 22
2	Mar 23
3	Sep 23

① Insert row in the Students table → Allows only if FK is present

② Delete a batch → **Orphaning**

→ deleting students of that batch

→ set the batch-id of those students to NULL

→ disallow removing that batch

③ Updating a batch → **inconsistent**

→ update the batch-id in students table too

→ set to NULL

→ disallow updates.

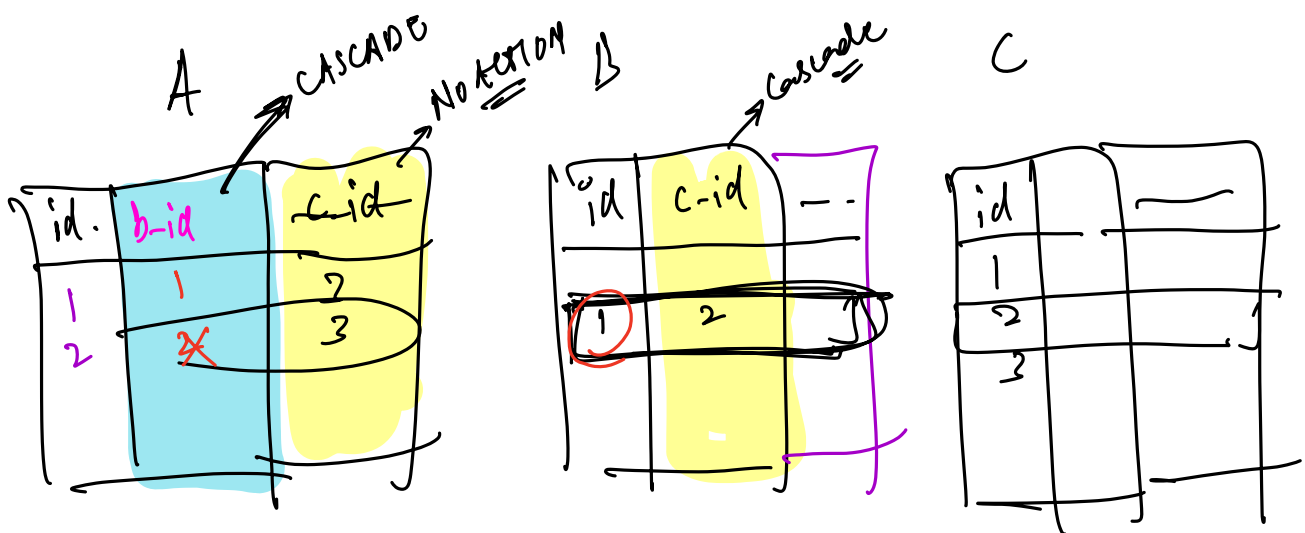
Ultimately there are 3 things that we can do:

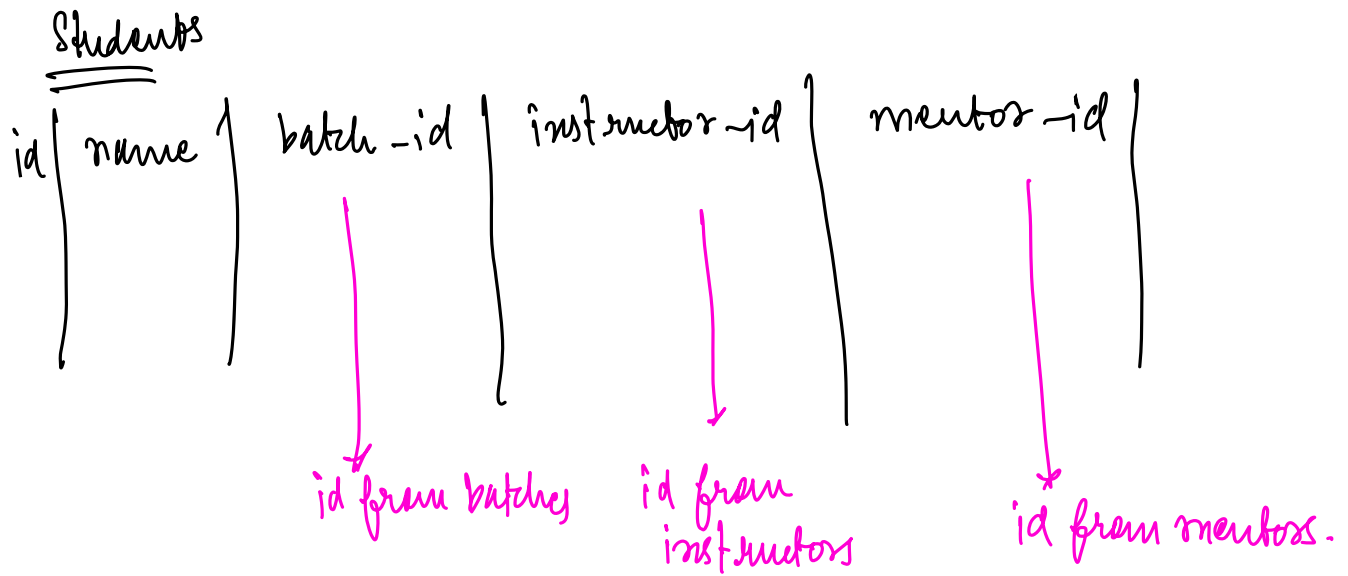
① Do what you did ⇒ **CASCADE**

② Set NULL ⇒ **SET NULL**

③ Don't Allow ⇒ **NO ACTION**

In SQL, when you create a FK, you can specify what to do in case of DELETE OR UPDATE operation on the other (foreign) table.





bookmarks

id	name	class-id
1	Agenda	2
2	Foreign key	2

FK (Example of cascade) (from class-id to class-id)

Classes

id	name
2	SQL: Key
2	SQL: CRUD
4	SQL: CRUD

Deleted (row with id 2, name SQL: Key)

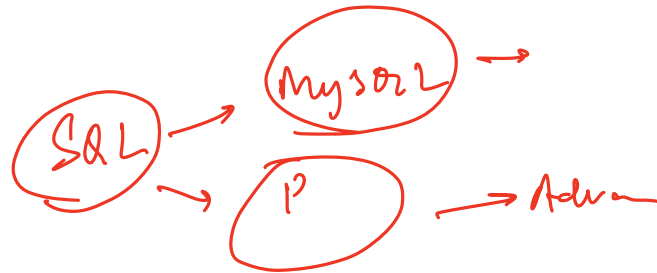
Homework

Install

- ① MySQL 8
- ② MySQL Workbench
- ③ Sakila Dataset

→ official MySQL Dataset

→



CRUD → Delete
↓
Create Read Update

Students

id	name	esp	batch-id
1	Alok	90	20 23
2	Mukul	91	21
3	Sushmita	92	<u>23</u>

- read data (R)
- update data (U)
- delete data (D)
- insert data (C)

CRUD

#create (Insert)

Syntax:-

```
Insert Into {table-name}
(col names)
VALUES
(           ),
(           ),
(           ) ;
```

keys →

CRUD

↓ → Read (The most important)
Done select statements

Please install MySQL